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Profitability: returns are low because the margin is low

Margins and returns on capital – formula, Liebherr and Caterpillar, 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
PROFITABILITY margins	Operating margin	EBIT / revenue	2.8 %	17.2 %
	Pre-tax margin	Profit before tax / revenue	3.0 %	17.9 %
	Net profit margin	Profit after tax / revenue	1.8 %	13.8 %
	EBITDA margin	EBITDA / revenue	6.8 %	19.8 %
PROFITABILITY returns on capital	ROA	EBIT / average total assets	2.4 %	11.9 %
	ROCE	EBIT / (avg. equity + avg. non-current debt)	3.5 %	20.2 %
	ROE	Profit after tax / average equity	3.0 %	38.1 %*
	Pre-tax ROE	Profit before tax / average equity	4.8 %	49.4 %

* Not comparable – treasury shares reduce Caterpillar's equity. ROA is measured on EBIT, because Liebherr's interest expense is not disclosed.

DUPONT

ROE = net margin × asset turnover × leverage

leverage = average total assets / average equity

= **profitability** × **efficiency** × **leverage** – the ratio groups of slides 14 and 15

	MARGIN	TURNOVER	LEVERAGE	ROE
2025	1.84 %	× 0.85	× 1.89	**3.0 %**
pre-tax	2.96 %	× 0.85	× 1.89	**4.8 %**
2023 pre-tax	4.76 %	× 0.88	× 1.81	**7.6 %**

Leverage works on ROE, **not on the EBIT margin** – EBIT is measured before interest.

Source: Excel workbook, sheets D3 (blocks C and D) and D4; Report §5.3, Table J14. 2025; Caterpillar on the FIFO basis

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Efficiency, leverage, liquidity: slow capital, solid finance

How capital is used, how it is financed and how liquid Liebherr is – 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
EFFICIENCY asset utilisation · working capital	Asset turnover	Revenue / average total assets	0.85	0.69
	Fixed asset turnover	Revenue / average PP&E	3.49	4.74
	Days inventory outstanding	Avg. inventories / revenue × 365	157.5	116.5
	Days sales outstanding	Avg. receivables / revenue × 365	73.8	54.5
	Days payables outstanding	Avg. payables / revenue × 365	28.6	44.9
	Cash conversion cycle	DIO + DSO - DPO	202.7	126.1
LEVERAGE solvency	Equity ratio	Equity / total assets	53.5 %	24.9 %
	Leverage (DuPont)	Avg. total assets / avg. equity	1.89	3.97
	Debt / equity	Debt incl. pensions / equity	0.40	1.84
	Debt / capital	Bank + lease liab. / (those + equity)	23.7 %	64.8 %
	Debt / EBITDA	Bank + lease liabilities / EBITDA	2.90	3.52
LIQUIDITY short term	Current ratio	Current assets / current liabilities	2.31	1.55
	Quick ratio	(Current assets - inventories) / current liab.	1.16	0.94
	Cash cover	Cash + current fin. assets / current debt	2.6×	–

READ WITH CARE

Days = efficiency, but they drive liquidity: 203 days of revenue must be financed.

Revenue basis: Liebherr reports no cost of sales; on its own cost basis DIO would be 299 days – not comparable.

Two debt measures: “debt” includes pensions (€3,714 m); bank and lease liabilities (€2,898 m) match the Damodaran benchmark.

Interest cover: not computable – no interest expense disclosed. Indicator: operating profit / interest paid 3.90.

PROFITABILITY
How much profit a company earns – on its revenue and on the capital it employs.

EFFICIENCY
How well a company uses its assets and working capital to generate revenue.

LEVERAGE
How strongly a company uses debt relative to its equity to finance its assets.

LIQUIDITY
Whether a company can pay its short-term obligations from its short-term assets.

Source: Excel workbook, sheets D3 (blocks D, E, G, H), D4 and D7; Report Tables J5 and J14. 2025; days on average balances; Caterpillar on the FIFO basis

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Cash flow and growth: profit does not turn into cash

Cash flow, payout and growth ratios – 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
CASH FLOW converting profit into cash	Cash conversion	Operating cash flow / EBITDA	53.0 %	87.5 %
	Capex / depreciation	Capex / depreciation and amortisation	1.20	1.90
	OCF / capex	Operating cash flow / capex	0.75	2.74
	Free cash flow	Operating cash flow - capex (€ m / $ m)	−176	7,453
	FCF margin	Free cash flow / revenue	−1.2 %	11.0 %
PAYOUT AND GROWTH what is left to grow with	Revenue growth	Revenue / prior-year revenue - 1	1.0 %	4.3 %
	Payout ratio	(Dividends + buy-backs) / profit after tax	20.6 %	85.2 %
	Sustainable growth rate	ROE × (1 - payout ratio)	2.4 %	5.7 %

READ WITH CARE

Rental fleet counted once: its investment sits inside operating cash flow, not in capex.

Two "cash conversions": OCF / EBITDA (53.0 %) is not the cash conversion *cycle* (202.7 days).

Sustainable growth 2.4 %: faster growth needs better returns, less capital or outside money.

Source: Excel workbook, sheets D5 (rows 20–25), D7 and D8 (rows 51–52); Report §5.3 and §5.6. 2025; Caterpillar capex incl. leased equipment

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From workbook to ratio: every figure is a formula

Excerpt of sheet D3 – inputs link to the recast statements, ratios link to the inputs

Liebherr-Caterpillar-Restructured-financial-statements.xlsx · sheet D3 Ratios

B40 | *fx* `=B10/B27`

	A	B · 2025	formula in column B
7	Revenue	14,772	`='B1 Income statement by nature'!B5`
8	Operating profit (EBIT)	413	`='C1 Accounting adjustments'!B30`
10	Profit after tax	272	`='B1 Income statement by nature'!B18+'C1…'!B27`
14	Total equity	9,354	`='C1 Accounting adjustments'!B21`
16	Inventories	6,341	`='C1 Accounting adjustments'!B13`
19	Current assets	12,709	`='B2 Restructured balance sheet'!B17+'C1…'!B12`
20	Current liabilities	5,498	`='B2 Restructured balance sheet'!B33`
27	Average total equity	9,128	`=(B14+C14)/2`
29	Average inventories	6,373	`=(B16+C16)/2`
35	Operating margin	2.80 %	`=B8/B7`
40	Return on equity (ROE)	**2.98 %**	`=B10/B27`
49	Days inventory outstanding	157.5	`=B29/B7*365`
60	Current ratio	2.31	`=B19/B20`

Return on equity `=B10/B27`

272 / 9,128 → **3.0 %**

Days inventory outstanding `=B29/B7*365`

6,373 / 14,772 × 365 → **157.5 days**

Current ratio `=B19/B20`

12,709 / 5,498 → **2.31**

0 open differences on sheet E2

Inputs come from the recast statements B1/B2 and the adjustments on C1 – nothing is typed in.

Source: Excel workbook, sheet D3 Ratios (cell formulas as in the file), column B = Liebherr 2025, € m; checks on sheet E2

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What-if I: direction matters more than the formula

Balance sheet and liquidity – what happens to the ratios

Suppliers paid in cash	Current ratio ▲ Cash ▼	Both sides fall by the same amount, so a ratio above 1 rises: €500 m paid → 2.31 to 2.44 (illustration).
Inventory write-down	EBIT ▼ Current ratio ▼ DIO ▼	The days **look** better while the position is weaker – the allowance is not disclosed.
New bank debt for capex	Equity ratio ▼ Debt/EBITDA ▲ Leverage ▲	EBIT margin unchanged; net margin falls through interest. ROE rises only if the return beats the interest rate.
Cash conversion cycle −10 days	Cash ▲ Margin –	Releases about €405 m once (€40.5 m revenue a day) – not a recurring improvement.
Receivables collected faster	DSO ▼ OCF ▲ Current ratio –	Cash replaces receivables inside current assets, so liquidity ratios barely move.

INVESTOR LENS **A better ratio is not always a better company**

Source: Report §5.4–5.5; Excel workbook, sheets D3 and D7. Directions are logical consequences; the €500 m case is an illustration on 2025 balances

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What-if II: accounting choices move profit, not cash

Effect of an accounting choice on EBIT, equity and cash

CHANGE	EBIT	EQUITY	CASH / OCF	LIEBHERR EXAMPLE
Provision addition	▼	▼	–	Expense now – the obligation is estimated
Provision usage (utilisation)	–	–	▼	Obligation settled – no profit effect
Provision reversal (release)	▲	▲	–	Estimate no longer needed – profit without cash
Slower depreciation (longer life)	▲	▲	–	Liebherr 5.5 % vs. Caterpillar 6.6 % of cost
More own work capitalised	▲ now	▲	–	€317 m = 77 % of 2025 EBIT; returns as depreciation
LIFO → FIFO, rising prices (CAT)	▲	▲	–	+$441 m EBIT, +$4,305 m equity before tax

INVESTOR LENS **Operating cash flow is the check on the accruals – only 28.4 % of EBITDA in 2022–2025**

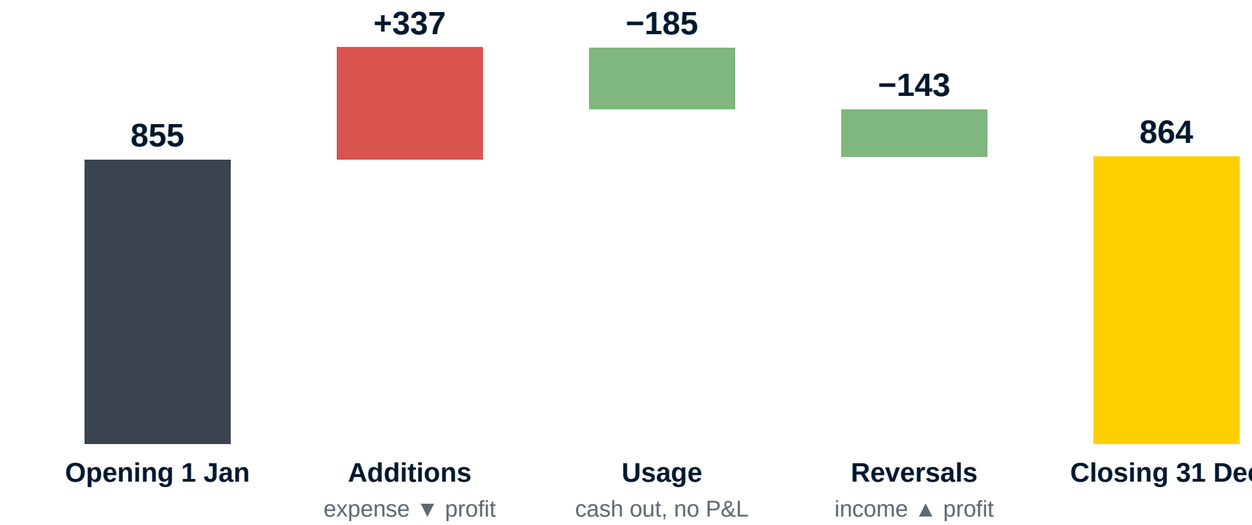
Source: Report §4.2–4.4 and §6.2; Excel workbook, sheets A2 (note 10), C1, C2 and D9

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Provisions: reversals rose – but so did additions

Movement of provisions 2025 (IAS 37), € m – of which warranty 530

Roll-forward 2025, all provisions



Net charge = additions − reversals	2022	2023	2024	2025
	193	338	195	194

How to read it

- **IAS 37:** present obligation from a past event, probable outflow, reliable estimate
- **Usage ≠ reversal:** only additions and reversals touch profit
- **Hidden reserve test:** a release to support profit would show as a *falling* net charge – 195 → 194, so not visible
- **Flag kept:** other provisions built up by €146 m in 2023 are being released (€63 m in 2025); the notes cannot show why

Source: Excel workbook, sheets A2 (note 10, rows 68–80), C2 and D11 (block B); Report §4.2 and §4.4, Table H1 test 1. € m

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Sensitivity: every measured effect lowers the margin

Liebherr operating margin 2025 – illustrations on published figures, not booked adjustments

STEP	CALCULATION, € M	EBIT	MARGIN
As reported	–	413	2.80 %
Net provision charge at the 2022–2024 average	413 - (1.76 % - 1.31 %) × 14,772	347	2.35 %
… without the high charge of 2023	413 - (1.43 % - 1.31 %) × 14,772	395	2.68 %
Caterpillar's depreciation rate (6.6 % instead of 5.5 % of cost)	413 - 105	308	2.08 %
Both effects together	413 - 66 - 105	242	1.64 %

2.80 − 0.44 − 0.71 = 1.64 %

Sheet D9 also shows a **gross-reversal** view (2.5 % at average reversals, 1.8 % without any reversal). The report uses the **net charge** (D11), because additions rose together with the reversals. Own work capitalised (€317 m) cannot be measured.

Source: Excel workbook, sheets D9 (blocks A and B) and D11 (block B); Report §6.1, Tables H3 and J13

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Accounting analysis in six steps

The framework of Palepu et al. (2019), applied to Liebherr

1	**Identify key accounting policies**	30 treatments from notes 2 to 11 (long list); key accounts follow from the key success factors – 7 accounts	Table G2, G1
2	**Assess accounting flexibility**	Judgement the standard permits × materiality – 9 items with high flexibility, plus revenue timing and the by-nature format	Table G2
3	**Evaluate accounting strategy**	Evidence points both ways – conservatism plausible, not proven (no capital-market pressure)	§4.3
4	**Evaluate the quality of disclosure**	The hardest limit – no tax reconciliation, pension assumptions, allowances, guarantees, related party note, note on other operating items (€2,572 m)	§4.3
5	**Identify red flags**	8 standard tests (Palepu) + 1 consistency check – 4 raised, 3 clean, 2 not testable	Table H1
6	**Undo distortions**	Short list of 22 findings – 7 not adjustable, 5 reversed in the workbook (e.g. LIFO → FIFO), 10 not reversible	Table H2, H3

Source: Report chapter 4 and Appendices G and H; Palepu et al. (2019), six steps of accounting analysis

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From 30 treatments to 22 findings

Four steps of the accounting analysis – the short list adds what the screen and the comparison found

STEP 1 · TABLE G2

Long list

30

accounting treatments with room for choice

▶

STEP 2 · TABLE G1

Key treatments

9

key account × high or medium-to-high flexibility

▶

STEP 3 · TABLE H1

Red-flag screen

9

8 tests + 1 consistency check → 4 flags · 3 clean · 2 not testable

▶

STEP 4 · TABLE H2

Short list

22

findings – red flags, gaps and comparability issues, not a subset of the 30

The nine key treatments

Inventory cost and write-down · own work capitalised · impairment · expected credit losses · three provision categories · deferred taxes. Revenue timing and the by-nature format are added – they affect every ratio.

Left out – and why

Borrowing costs, lessee accounting and currency translation: the standard prescribes the mechanics. Development costs: never capitalised, kept only for the comparison. 7 key accounts hold 9 key treatments; the 9 tests are a different count.

Source: Report §4.2–4.4, Tables G1, G2, H1 and H2

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Long list I: revenue and operating assets

Treatments 1–18 of 30 (Table G2) – where Liebherr's accounts leave room for choice

#	ACCOUNT AND POLICY (STANDARD)	FLEXIBILITY	WHY	KEY ACCOUNT?
A · REVENUE AND CUSTOMER CONTRACTS				
A1	Timing of revenue (IFRS 15)	Medium	cost-to-cost estimate, over-time share not disclosed	Yes
A2	Sale of rental machines (IAS 16.68A)	Medium	residual value, amount not disclosed	Partly
A3	Leases as lessor (IFRS 16)	Medium	risks and rewards judgement, small balance	No
A4	Sales with repurchase obligations (IFRS 15)	Low	rule-based test, liability 32	No
A5	Financial guarantees for customer financing (IFRS 9)	Medium	allowance judgement, volume not disclosed	Gap
A6	Advance payments and contract liabilities (IFRS 15)	Low–medium	balances 896 and 289	Partly
B · OPERATING ASSETS				
B1	Cost of inventories (IAS 2)	High	normal capacity estimate, inventories 6,341	Yes
B2	Write-down of inventories (IAS 2)	High	undisclosed allowance on 6,341	Yes
B3	Own work capitalised (IAS 16, IAS 38)	Medium–high	attributable cost, 77 % of EBIT	Yes
B4	Development costs (IAS 38.57)	Medium	IAS 38 criteria, R&D 708	Peer only
B5	Useful lives and residual values (IAS 16)	Medium	bounded by disclosed ranges, on 4,304	Partly
B6	Impairment of non-current assets (IAS 36)	High	indicator and cash flow judgements	Yes
B7	Subsequent expenditure on PP&E (IAS 16.13)	Medium	undefined threshold, additions 1,050	Partly
B8	Borrowing costs (IAS 23)	Low	rule-based, 14 capitalised	No
B9	Goodwill (IFRS 3, IAS 36)	Low	immaterial, intangibles 29	No
B10	Leases as lessee (IFRS 16)	Low	largely rule-based, liability 238	No
B11	Expected credit losses on receivables (IFRS 9)	Medium	provision matrix, allowance not disclosed	Yes
B12	Derecognition of receivables (IFRS 9)	Low–medium	rule-based, no sale disclosed	No

Source: Report Appendix G, Table G2 (30 treatments, notes of the 2025 annual report); flexibility = judgement the standard permits × materiality

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Long list II: provisions, tax and presentation

Treatments 19–30 of 30 (Table G2) – “Yes” = key treatment carried into section 4.2

#	ACCOUNT AND POLICY (STANDARD)	FLEXIBILITY	WHY	KEY ACCOUNT?
C · PROVISIONS AND EMPLOYEE BENEFITS				
C1	Warranty provisions (IAS 37)	High	failure-rate estimate, provision 530	Yes
C2	Expected losses from pending transactions (IAS 37)	High	contract cost estimate, linked to A1	Yes
C3	Other provisions and product liability (IAS 37)	High	content undefined, reversals 63	Yes
C4	Restructuring provisions (IAS 37)	Low–medium	rule-based trigger, none disclosed	No
C5	Post-employment benefits (IAS 19)	Medium	actuarial estimates, 791	Partly
D · TAX AND FINANCIAL ITEMS				
D1	Deferred tax assets and tax expense (IAS 12)	High	probability judgement, swings the tax charge	Yes
D2	Securities at fair value (IFRS 9, IFRS 13)	Low	level 1 and 2 inputs, 1,568	Partly
D3	Derivatives and hedge accounting (IFRS 9)	Medium	designation choice, movement 160	Partly
D4	Foreign currency and hyperinflation (IAS 21, IAS 29)	Low	rule-based, effect not quantified	No
E · SCOPE AND PRESENTATION				
E1	Scope of consolidation (IFRS 10, 11, IAS 28)	Low	control judgement, equity-accounted 2	No
E2	Income statement by nature (IAS 1)	Medium	line allocation, other operating expenses 2,170	Yes
E3	Presentation of the finance result (IAS 1, IFRS 7)	Medium	finance lines not split	Partly

Source: Report Appendix G, Table G2 (30 treatments, notes of the 2025 annual report); flexibility = judgement the standard permits × materiality

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Seven key accounts carry the judgement

Key accounts × flexibility = the nine key treatments of the long list

KEY ACCOUNT	2025, € M	STANDARD AND JUDGEMENT	FLEXIBILITY	WHY IT IS ON THE SHORT LIST
Inventories long list B1, B2	**6,341** 42.9 % of revenue	**IAS 2** – Fixed overheads absorbed at an undefined normal capacity; allowance not disclosed	High	Largest account – 36.3 % of total assets; quality cannot be verified
Provisions long list C1–C3	**864** of which warranty 530	**IAS 37** – Recognition, measurement and reversal of the estimate	High	Reversals €143 m = 34.6 % of the operating result if credited within it
PP&E and rental fleet long list B5–B7	**4,304** 24.6 % of total assets	**IAS 16 / IAS 36** – Useful lives, residual values, indicator-based impairment	High (impairment), medium (lives)	Depreciation 5.5 % of cost vs. 6.6 % at Caterpillar
Own work capitalised long list B3	**317** 77 % of the operating result	**IAS 16 / IAS 38** – Which production cost is capitalised rather than expensed	Medium to high	Moves cost out of the year – returns later as depreciation
Trade receivables and contract assets long list B11	**3,116** 17.8 % of total assets	**IFRS 9** – Simplified lifetime expected credit loss through a provision matrix	Medium	Grew 9.1 % against revenue +1.0 % in 2025; allowance not disclosed
Revenue long list A1, A2	**14,772**	**IFRS 15** – Point in time against over time; cost-to-cost estimate	Medium	Share recognised over time not disclosed – affects every ratio
Deferred tax assets and tax charge long list D1	**285** tax rate 37.8–99.2 %	**IAS 12** – Probability of future taxable profit; announced instead of enacted rates	High	No tax reconciliation – after-tax ratios not usable

Source: Report §4.2–4.3, Tables G1 and G2; Liebherr-International AG (2026); IFRS Foundation (n.d.)

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Red-flag screen: nine tests, four flags

4 raised · 3 clean · 2 not testable – reasons to ask, not proof of manipulation

#	TEST	EVIDENCE 2022–2025	VERDICT
1	**Unexplained transactions that boost profits**	Provision reversals 90 → 143 (2022–2025) = 34.6 % of 2025 EBIT; net charge flat (195 → 194); credited line not disclosed	Raised
2	**Inventories or receivables rise faster than sales**	Inventories +27.3 % vs. revenue +8.2 % (2022); receivables +9.1 % vs. revenue +1.0 % (2025)	Raised twice
3	**Gap between profit and cash flow or taxable profit**	Operating cash flow −415 and −10 (2022–23) against profit; tax rate 99.2 % → 37.8 %, no reconciliation	Raised
4	**R&D partnerships, special purpose entities, sold receivables**	None disclosed in four years; R&D fully expensed	No flag
5	**Unexpected large asset write-offs**	PP&E impairment 8, 55, 9, 3 – 2023 six times the other years, same one-sentence explanation	Raised for 2023
6	**Large year-end adjustments**	No interim figures published, so no within-year comparison	Not testable
7	**Qualified opinions or auditor changes**	Ernst & Young all four years, no qualification, no restatement (ISA 810 opinion)	No flag
8	**Related party transactions**	No note in four years – family-owned group that also runs six hotels	Not testable
9	**Changes in accounting policies or estimates**	Policies worded identically; new standards without material effect	No flag

Source: Report §4.4, Table H1 (8 tests of step 5 + consistency check of step 3); Liebherr-International AG (2023–2026)

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Short list I: what cannot be fixed – and what we fixed

Findings 1–12 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT ADJUSTABLE – RED FLAGS AND DISCLOSURE GAPS · 7			
1	**Provision reversals rise as the result falls (test 1)**	Provisions, 864	Not adjustable: the income line carrying the reversal is not disclosed
2	**Inventories and receivables grow faster than sales (test 2)**	Inventories 6,341, receivables 3,116	Not adjustable: neither allowance is disclosed
3	**Volatile and unreconciled tax charge (test 3)**	Deferred tax assets, 285 (IAS 12)	Not adjustable: no tax reconciliation is published
4	**Impairment spike in 2023 (test 5)**	PP&E, impairment 55	Not adjustable: no test assumptions are disclosed
5	**Composition of other operating expenses not disclosed**	Other operating expenses, 2,170	Not adjustable: the line has no note
6	**No related party disclosure in four years (test 8)**	Whole group	The information is not in the published statements
7	**Customer credit risk carried off balance sheet**	Financial guarantees, not quantified	Not adjustable: no amount is disclosed
REVERSED IN THE WORKBOOK · 5			
8	**Caterpillar inventories on LIFO**	Inventories, equity, operating profit	FIFO excess disclosed – exact correction: $4,305 m
9	**Caterpillar goodwill impairment 2022 inside operating costs**	Operating profit 2022	Moved to other income as a non-recurring item
10	**Liebherr change in inventories and own work shown as income**	Revenue, cost of materials	Both mapped into the cost line (standardised format)
11	**Caterpillar finance receivables inside trade receivables**	Trade receivables	Mapped to other current assets and non-operating investments
12	**Liebherr trade payables not shown on the balance sheet**	Trade payables	Taken from note 7, published subtotal preserved

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

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Short list II: what the disclosure does not let us reverse

Findings 13–22 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT REVERSIBLE – DISCLOSURE TOO THIN · 10			
13	**Gross presentation of rental machine disposals**	Revenue, cost of materials	The amount is not disclosed
14	**The interest element of the finance result**	Finance income 1,043, expenses 1,015	No split is published
15	**No segment note under IFRS 8**	Whole income statement	Segment data only in the management report
16	**The effect of hyperinflation accounting**	Turkish and Ghanaian entities	Not quantified in any year
17	**Actuarial assumptions behind the pension obligation**	Post-employment benefits, 791	No discount rate and no sensitivity disclosed
18	**Inventory and receivable allowances**	Inventories 6,341, receivables 3,116	Neither balance nor charge is disclosed
19	**The two groups do not depreciate at the same rate**	PP&E, 4,304	Measured (Table H3), not carried into the recast – asset mixes differ
20	**Cat Financial inside every Caterpillar figure**	Caterpillar balance sheet	The consolidated basis was set by the assignment
21	**Caterpillar discloses no cost of materials or personnel expense**	Cost structure comparison	Not estimated – operating costs shown as one line
22	**Potential refund of US tariffs after the balance sheet date**	Not recognised; note 12	Non-adjusting event (IAS 10), no reliable estimate

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

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Workbook: every number traces back to the report

From the published statements to the ratios – and how we know it adds up

A1–A4

Published statements

typed exactly as printed

B1–B3

Recast by nature

formulas only · 21 mapping decisions

C1–C2

Adjustments

LIFO → FIFO · goodwill 2022 · note data

D1–D11

Analysis

ratios · DuPont · sensitivities · benchmark

E1–E2

Judgements and checks

44 judgements · every subtotal reconciled

Key mapping decisions

- Change in inventories and own work capitalised mapped into the cost of materials
- Caterpillar's finance receivables taken out of trade receivables
- Liebherr's trade payables taken from note 7
- Caterpillar's operating costs kept as one line – no materials or personnel split published
- Liebherr in €, Caterpillar in $ – never added or divided across companies

0

CHECKS · SHEET E2

open differences

Three deliberate differences are labelled (goodwill 2022, affiliates after tax, depreciation)

WHY BY NATURE, NOT BY FUNCTION

Liebherr reports by nature (materials, personnel, depreciation), Caterpillar by function (cost of sales, SG&A, R&D). A function format would leave every Liebherr cost line empty and show no depreciation for either group – so both are recast by nature.

Source: Excel workbook, sheets Method, B1–B3, C1, E1 and E2; Report §4.5 and Appendix K

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Environment: a modest recovery, but costs rise first

DESTEP scan

Economic

European construction equipment market +4.6 % in 2025, +2 to 2.5 % expected for 2026 – while Liebherr's construction and mining revenue fell 5.5 %.

Political

US tariffs cut Liebherr's North American crane revenue in 2025. Caterpillar expected a tariff impact of $2,600 m in 2026.

Social

Construction had 792 fatal accidents at work in the EU in 2023, 24.0 % of the total. Safety becomes a purchase criterion.

Technological

Autonomous and electrified machines. EU Regulation 2023/1230 covers autonomous functions from 20 January 2027.

Ecological

EU emission limits for non-road machinery and the target of at least 55 % net emission cuts by 2030 make low-emission machines a purchase criterion.

Demographic

Shortages of skilled workers and machine operators push costs up. Personnel expenses reached 28.6 % of revenue in 2025.

Source: Report §3.1.1; CECE (2026), Eurostat (2025), European Union (2023), European Commission (n.d.-a, n.d.-b), Liebherr-International AG (2026), Caterpillar Inc. (2026)

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Competitive forces squeeze the margin

Porter's five forces

New entrants	Moderate, rising	Chinese producers enter with cheaper, simplified machines. Mobile and crawler crane revenue fell 6.8 % in 2025.
Supplier power	Moderate, rising	Cost of materials is 52.7 % of revenue. Electrification shifts power to battery and electronics suppliers.
Buyer power	High, stable	Receivables rose 9.1 % against revenue growth of 1.0 % in 2025. About half of fleet owners may reconsider their brands by 2028.
Substitutes	Low to moderate, rising	Renting and automation change how much equipment is needed. The rental fleet took €305 m, 28.8 % of 2025 investment.
Rivalry	High, rising	The fixed cost of staying in the market rises. The operating margin fell from 4.6 % in 2024 to 2.8 % in 2025.

INVESTOR LENS **Limited industry profit potential – consistent with the fall to 2.8 %**

Source: Report §3.1.2, Table 1; ratings are the judgement of the report

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Value chain: built for differentiation, paid for with capital

Porter's value chain – where Liebherr creates value and what it ties up

Upstream · production

- Materials 52.7 % + personnel 28.6 % = **81.3 % of revenue** – integration buys control, not low cost
- PP&E 29.1 % of revenue (Caterpillar 22.4 %)
- Investment €1,059 m = 1.8× depreciation (incl. €305 m rental fleet)
- Inventories 42.9 % of revenue; work in progress 19.7 % of stock (CAT 8.8 %)

Downstream · sales and service

- Own distribution and sales companies – Caterpillar: about 150 independent dealers in 190 countries
- So working capital and distribution cost sit on Liebherr's books
- Receivables 77 days of sales at year end (CAT 59)
- Other operating expenses 13.3 % → 14.7 % of revenue

Support activities

- Technology development 4.8 % of revenue (CAT 3.2 %), expensed in full – intangibles only €29 m
- Components supply drive and control technology to the machine segments
- Internal transfers not quantifiable – no segment data

INVESTOR LENS **The value chain explains the asset base – and the thin margin**

Source: Report §3.2.1 (value chain after Porter, 1985), Appendices B and C; Liebherr-International AG (2026); Caterpillar Inc. (2026)

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Hard-to-copy resources – locked in the asset base

VRIO and SWOT

VRIO: what is hard to copy

- **Sustained advantage:** in-house key technologies (40+ production sites in 16 countries), application engineering (R&D €708 m, 4.8 % of revenue), family ownership with permanent capital (equity ratio 53.5 %)
- **Parity only:** wholly-owned sales and service network, decentralised divisional structure
- **Temporary to sustained:** internal capital market

Strengths

Vertical integration · equity ratio 53.5 % · 13 segments

Weaknesses

Margin 2.8 % vs. 17.2 % at Caterpillar · operating cash flow €1,171 m → €530 m · no segment disclosure

Opportunities

Low-emission and automated machines · European market +4.6 % in 2025

Threats

Low-price competitors from China · US tariffs · transition costs for new drives and autonomy

Source: Report §3.2 and §3.4, Tables E1 and D1

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Corporate strategy: patient capital, one real synergy

Is the group worth more than its 13 segments? Three market imperfections, decision rights, information

Financial market	Strongest	Internal capital market – invested €1,059 m vs. depreciation €587 m; only €56 m of a €272 m profit distributed; R&D 588 → 708 while EBIT fell 39 %. Depends on internal funding: OCF €1,171 m → €530 m
Product market	Explains the assets	Components for extreme applications are not off the shelf – PP&E and inventories are 60.9 % of total assets
Labour market	Explains costs	Own training – 3,186 apprentices and interns – because skilled trades are short
Decision rights	A governance issue	Holding only coordinates; decentralised divisions; board of family members only – no capital-market scrutiny
Information systems	Weakest	No internal key figures, no remuneration report, no segment note – capital allocation cannot be assessed

What holds it together

Shared components are the one genuine operating synergy.

Where a segment shares neither components nor customers – hotels, domestic appliances – the case rests on diversification alone.

Diversification stabilised revenue in 2025, but it does not raise the margin.

Source: Report §3.3 (corporate strategy after Palepu et al., 2019); Liebherr-International AG (2023–2026); European Labour Authority (2026)

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Red-flag screen: reasons to ask, not proof

Nine tests over four years

Four red flags

- **Provision reversals:** €143 m in 2025, 34.6 % of the operating result if credited within it
- **Receivables vs. revenue:** receivables +9.1 % against revenue +1.0 % in 2025
- **Profit vs. tax:** effective tax rate 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), no reconciliation
- **Impairment:** €55 m in 2023, six times the level of the other years

Clean or not testable

- **Clean:** no off-balance-sheet arrangements disclosed · same auditor (Ernst & Young), no qualified opinion · no change of policy or estimate
- **Not testable:** related party transactions (no note in four years) · large year-end adjustments (no interim figures)

4
flags raised

3
clean

2
not testable

Source: Report §4.4, Table H1; financial statements 2022–2025

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Where management has room for judgement

Rated by the judgement the standard allows and the materiality of the amount

Highest flexibility

- Inventories: €6,341 m, 36.3 % of total assets, allowance not disclosed
- Provisions: €864 m, of which €530 m warranty
- Impairment test
- Deferred tax assets: €285 m

Also material

- Own work capitalised: €317 m, 77 % of the operating result
- PP&E: €4,304 m, machinery written off over 5 to 21 years
- Revenue timing: advance payments €896 m, contract liabilities €289 m

Lowest flexibility

- Borrowing costs
- Lessee accounting
- Translation of foreign currency
- The standard prescribes the mechanics – little choice for management

Source: Report §4.2–4.3, Table G2 (30 treatments, nine of them key)

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Conservative or aggressive? Both, in parts

Step 3 of the accounting analysis – the evidence points both ways

Points to conservative

- Development costs never capitalised (IAS 38.57)
- Policies worded identically in all four years
- Incentives to inflate profit are absent: unlisted, family-owned, no share-based pay, no disclosed covenant
- Warranty provision 3.6 % of revenue – assumptions not disclosed, so not provable

Points to aggressive

- Own work capitalised €317 m = 77 % of the operating result
- Provision reversals €143 m = 34.6 % of the operating result
- Deferred tax assets €285 m, with announced instead of enacted tax rates
- Slower depreciation than Caterpillar (5.5 % vs. 6.6 % of cost)

OUR VERDICT

Conservatism is plausible, but not proven

Palepu: conservative accounting can mislead too – reserves built in good years can be released in weak ones.

That fits other provisions built up in 2023 (+ €146 m) and released in 2025 (€63 m).

INVESTOR LENS **The measurable effects make 2025 look better: margin 1.6–2.8 % instead of 2.8 %**

Source: Report §4.3 (accounting strategy) and §6.1–6.3; Palepu et al. (2019); Liebherr-International AG (2026)

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22 findings: what we could adjust and what we could not

Short list of the accounting analysis

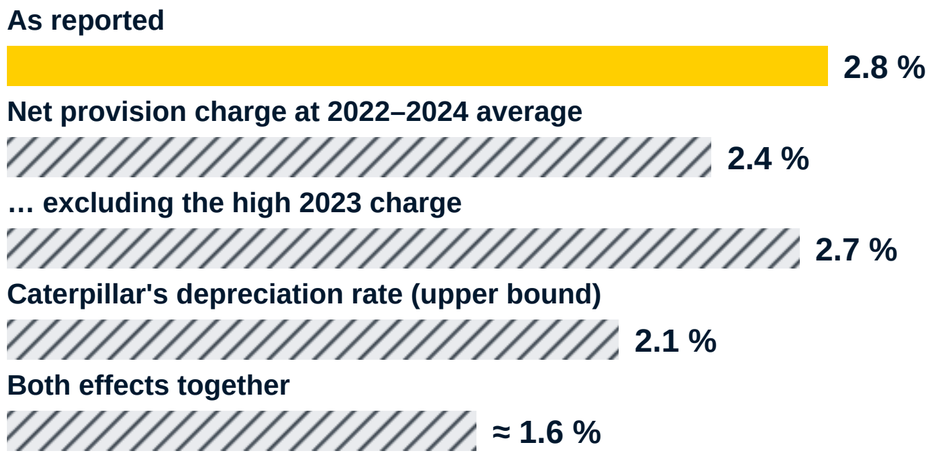
7 **not adjustable**
Red flags and disclosure gaps, e.g. provision reversals of €143 m, missing related party note

5 **reversed in the workbook**
e.g. Caterpillar LIFO → FIFO: equity +$4,305 m, operating profit +$441 m; goodwill impairment 2022

10 **not reversible**
Disclosure too thin, e.g. undefined normal capacity, own work capitalised €317 m

ILLUSTRATIVE SENSITIVITY · NOT BOOKED

Liebherr operating margin 2025



The quantified effects point to a weaker result – the weakness is, if anything, understated.

Source: Report chapter 4, Tables H2, H3 and J13; Excel workbook, sheets C1, D9 and D11

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How far the key ratios can be trusted

Table J13 by ratio group – what the accounting analysis does to each ratio

GROUP	RATIO 2025	FINDING OF CHAPTER 4	DIRECTION	SIZE WHERE MEASURABLE	USE
PROFITABILITY	**Operating margin 2.8 %**	Reversals 143, slower depreciation, own work 317, other expenses without note	Overstated	1.6–2.8 % in the scenarios	With bias
	Pre-tax margin 3.0 %	Finance result not split; fair value gains 55 = 12.6 % of profit before tax	Mixed	–	With bias
	Net margin 1.8 %, ROE 3.0 %	Tax rate 37.8–99.2 %, no reconciliation	Volatile	after tax +0.08 pp vs. pre-tax −2.14 pp	Not usable
EFFICIENCY	**DIO 157.5**	Overhead at undefined normal capacity, allowance not disclosed	Value possibly overstated	not measurable	Reliable
	DSO 73.8	Contract assets inside, allowance and guarantees not disclosed	Exposure understated	not measurable	With bias
	DPO 28.6	Trade payables taken from note 7	None	–	Reliable
	Fixed asset turnover 3.49	Depreciation 5.5 % vs. 6.6 % at Caterpillar	Understated vs. CAT	4.76 at CAT's rate (CAT 4.74)	With bias
LEVERAGE	**Equity ratio 53.5 %, D/E 0.40**	Pension assumptions, guarantees, depreciation	Slightly overstated	50.4 % at CAT's rate	With bias
	Interest cover	Finance result without interest component	Not computable	indicator 3.90 (2022: 11.07)	Not usable
LIQUIDITY	**Current 2.31, quick 1.16**	Securities 1,568 at fair value; Cat Financial at CAT	Moves with markets	fair value loss 209 in 2022	Reliable

Source: Report §6.3, Table J13; Excel workbook, sheets C1, C2, D3–D7, D9 and D11. Liebherr 2025, € m

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Growth and cost structure 2022–2025

Operating margin per year, with revenue and operating profit

2022
3.6 %
Revenue €12,589 m
Operating profit €454 m

2023
4.4 %
Revenue €14,042 m
Operating profit €619 m

2024
4.6 %
Revenue €14,622 m
Operating profit €674 m

2025
2.8 %
Revenue €14,772 m
Operating profit €413 m

Where growth stalled

- Revenue growth 11.5 % (2023) → 4.1 % (2024) → 1.0 % (2025)
- Construction and mining −5.5 % to €9,345 m
- Other segments +14.7 % to €5,427 m, components +35.5 %

Where the margin went

- Personnel expense +5.3 % to €4,226 m vs. revenue +1.0 %
- Personnel share of revenue 26.4 % (2023) → 28.6 % (2025)
- Other operating items cost another 1.25 pp
- Each margin point = €148 m operating profit

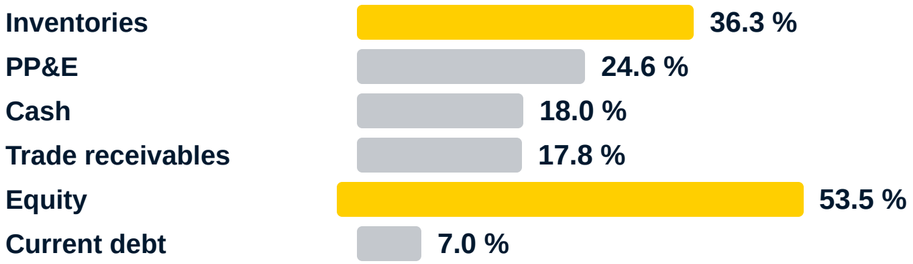
Source: Report §5.2; Excel workbook, sheets D1 and D6

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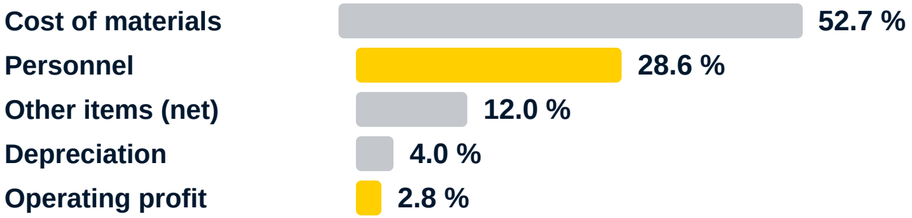
Common size: stock and staff dominate, profit is a sliver

Vertical analysis 2025 (balance sheet in % of total assets, P&L in % of revenue) and horizontal change 2024 → 2025

Balance sheet, % of total assets



Income statement, % of revenue



HORIZONTAL 2024 → 2025

Personnel expense	+5.3 %
Operating profit	−38.7 %
Trade receivables	+9.1 %
Current debt	+26.3 %
Inventories	−1.0 %
Revenue	+1.0 %

Why we recast

Published, the cost of materials fell from **60.0 %** of revenue (2022) to **54.6 %** (2025) – as if vertical integration paid off.

Recast – with the change in inventories and own work capitalised in the cost line – it stayed at **52.7–53.2 %**.

The “saving” was the stock build of 2022 and 2023.

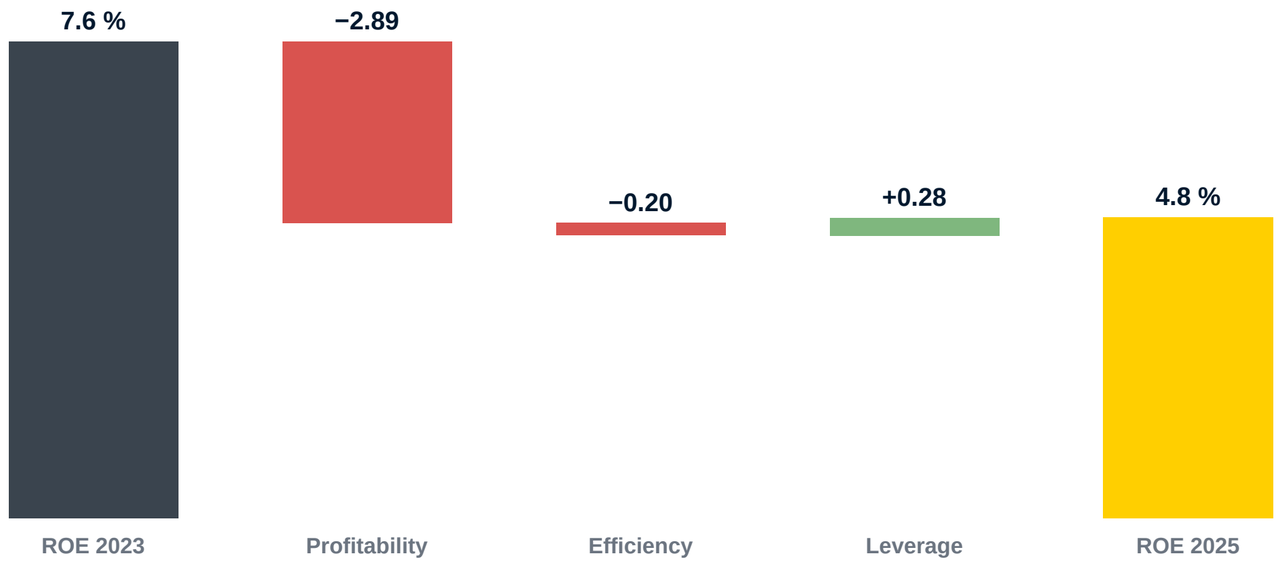
Source: Excel workbook, sheets D1 (vertical and horizontal analysis) and D2; Report §5.2 and §6.1. Liebherr, recast statements

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DuPont: the margin, not leverage, drives returns down

Pre-tax return on equity 2023 → 2025

Contributions to the change in pre-tax ROE, percentage points



pre-tax return on equity · contributions in pp

What it means

- **Profitability −2.89 pp:** lower pre-tax margin
- **Efficiency −0.20 pp:** slightly lower asset turnover
- **Leverage +0.28 pp:** more debt even helped a little
- After tax, ROE was 3.0 % vs. 16.4 % in the US industry – but the tax charge is volatile, so the trend is read before tax

Source: Report §5.3, Figure 2a, Table J4; Excel workbook, sheets D4 and D6 (logarithmic split)

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Strong solvency, far weaker returns

Liquidity, solvency and benchmark 2025

Balance sheet strength

- Equity ratio 53.5 %; bank and lease liabilities 23.7 % of capital (industry 39.3 %)
- Cash and current financial assets €3,147 m cover current debt of €1,215 m 2.6×
- Current ratio 2.31 vs. 1.55 at Caterpillar
- Net financial position €523 m (2022) → net debt €85 m (2023) → €249 m (2025)
- Debt / EBITDA 2.14 → 2.90; operating profit / interest paid (indicator) 11.07 (2022) → 3.90 (2025)

2025	LIEBHERR	US INDUSTRY	CATERPILLAR
Operating margin	**2.8 %**	15.9 %	17.2 %
EBITDA margin	**6.8 %**	19.6 %	19.8 %
Return on equity	**3.0 %**	16.4 %	38.1 %*
Inventories, % of revenue	**42.9 %**	16.6 %	33.2 %
Debt / EBITDA	**2.90**	2.30	3.52

* Not comparable – Caterpillar's equity is reduced by treasury shares. Caterpillar on the FIFO basis.

Source: Report §5.1 and §5.5, Table J14; Damodaran (2026b, 2026c, 2026d, 2026e), US machinery industry, January 2026; Excel workbook, sheets D3, D7 and D10

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Liebherr invests ahead of its cash

Cash flow and capital allocation

190.2 %
Capex in % of operating cash flow, 2022–2025

–€1,151 m
Free cash flow in total, 2022–2025

28.4 %
of EBITDA became operating cash flow, 2022–2025 (53.0 % in 2025)

€530 m
Operating cash flow 2025, down from €1,171 m (2024)

Capex above operating cash flow

- Capex and R&D 9.6 % of revenue vs. 7.4 % at Caterpillar – at a margin of 2.8 % vs. 17.2 %
- Shortfall after distributions €388 m (2023–2025)
- Bank and lease liabilities +€562 m (2022–2025)

Caterpillar returns cash

- Free cash flow $7,453 m in 2025, 11.0 % of revenue
- 73.1 % of operating cash flow returned to shareholders (2022–2025)
- Liebherr distributed €56 m in 2025 – little current income for an investor

Source: Report §5.6, Figure 4; Excel workbook, sheets D5, D7 and D10

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Why 3.9 % is only a trigger, not a target

Condition 1 of the reassessment

3.9 %
Average operating margin 2022–2025 (3.86 %)

+€157 m
Operating profit at 2025 revenue (€14,772 m)

3.5 %
ROCE before tax 2025 vs. 7.70 % after-tax benchmark

6–8 %
Operating margin needed to earn the benchmark

What the trigger means

- A return to 3.9 % is the first realistic sign of recovery – Liebherr has earned it on average
- It is a test, not a target return
- Each margin point is worth €148 m

Why it is not enough

- Even in 2024, at a 4.6 % margin, ROCE was only 5.9 % before tax – below the 7.70 % benchmark
- The benchmark is the US machinery industry, not Liebherr's own cost of capital
- A margin recovery alone does not prove an adequate return

Source: Report §5.3 and §5.7, chapter 8; Excel workbook, sheets D3 and D10; Damodaran (2026a)

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What the scenarios are worth

Operating profit at 2025 revenue (€14,772 m) – illustrations, not forecasts

Margin scenarios

2025 actual · 2.80 % margin



€413 m

Four-year average · 3.86 % margin



€570 m +€157 m

Best year 2024 · 4.61 % margin



€681 m +€268 m

US industry average · 15.86 % margin

€2,343 m +€1,930 m

The industry figure shows the size of the gap and is not a target.

Cash scenarios (2025)

- Free cash flow −€176 m: the net financial position of €249 m covers it 1.4 years
- After distributions (−€232 m): 1.1 years
- Capex at the level of depreciation (€587 m): +€119 m, free cash flow still −€57 m
- Cash conversion cycle 10 days shorter: €405 m released once

Source: Report Table J15; Excel workbook, sheet D10. Margins Liebherr has earned or the industry average – nothing is assumed

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Risks, limits and when to drop the case

What could still go wrong – and the alternative for a cautious investor

Demand

Construction and mining revenue −5.5 % in 2025; no significant upswing expected before 2027

Costs

US tariffs and the cost of autonomous and electrified machines press on the margin

Information

The largest cost increase of 2025 cannot be explained from the summary statements

Valuation

Not assessed in the report – price and contract rights decide the case

Alternative for a risk-averse investor

Equity is unsuitable: low payout, shares not traded. A loan may fit better, as equity of 53.5 % limits the default risk – but pricing, maturity, collateral and covenants are not disclosed.

Drop trigger

If the margin stays near 2.8 % in 2026 **and** the net financial position turns negative, the investor should drop the case rather than wait for 2027. A recovery should not rest on large provision releases.

Source: Report §5.5, §5.7, §6.1 and chapter 8

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Three levers and three changes to monitor

Chapter 7: what drives future performance · chapter 8: what Liebherr would have to change

1 Demand

Construction and mining −5.5 % in 2025; European construction about +2 % a year to 2028; no significant upswing before 2027

Threats: US tariffs, customer financing costs, low-price competitors

2 Margin

Each percentage point = €148 m of operating profit at 2025 revenue

Threats: tariffs, cost of autonomous and electrified machines

3 Working capital

10 days less cash conversion cycle = about €405 m released

But only once – lasting free cash flow needs margin

THREE CHANGES TO MONITOR IN 2026

- **Personnel costs follow orders and revenue** – a falling personnel ratio (28.6 % → towards 26.4 %)
- **Capex prioritised by expected return** and within operating cash flow over time – not simply cut
- **A shorter cash conversion cycle** – but it releases cash once, so lasting free cash flow also needs the margin

Due diligence

Request **segment results and assets** – without them capital allocation across 13 segments cannot be assessed.

Source: Report chapter 7 (three levers) and chapter 8 (changes to monitor, due diligence), sections 3.2.1, 3.3.3, 4.1, 5.2–5.7

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What to check in the 2026 statements

Table 3 – three conditions for a reassessment and seven supporting signals (not automatic triggers)

INDICATOR	2025	THRESHOLD OR SIGNAL	TYPE
Operating margin	**2.8 %**	At least 3.9 %, without large provision releases	Condition
Free cash flow	**−€176 m**	Positive and, over time, covering distributions	Condition
Information	**Summary statements only**	Audited statements and composition of other operating expenses	Condition
Personnel expenses / revenue	**28.6 %**	Falling back towards 26.4 % (2023)	Signal
Other operating expenses / revenue	**14.7 %**	Back towards 13.3 % (2024), composition disclosed	Signal
Capex / operating cash flow	**190.2 % (2022–2025)**	Below 100 % over a rolling four-year period	Signal
Cash conversion cycle	**202.7 days**	At least 10 days shorter as a first step	Signal
Net financial position	**€249 m**	Stays positive	Signal
Bank and lease liabilities / EBITDA	**2.90×**	Back towards the industry's 2.30×	Signal
ROCE before tax	**3.5 %**	Rising towards 7.70 % – needs a margin of about 6–8 %	Signal

Source: Report chapter 8, Table 3; Excel workbook, sheets C2, D1, D3, D5–D8, D10 and D11; Damodaran (2026a, 2026b). Thresholds are the judgement of the report

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Why not invest now?

Likely questions (1/4)

Q Why not invest now?

Neither financial condition was met in 2025: the margin of 2.8 % is below its four-year average of 3.9 %, and free cash flow was −€176 m. The answer is **not yet**, rather than no.

Q Why is 3.9 % the first test?

It is the average Liebherr earned from 2022 to 2025 – a realistic first sign of recovery. At 2025 revenue it adds €157 m of operating profit. A trigger, not a target return.

Q Is the LIFO to FIFO adjustment fair?

IAS 2 does not permit LIFO, so FIFO puts both groups on one basis. Caterpillar's LIFO reserve of $4,305 m lifts its margin only from 16.50 % to 17.15 % – the gap stays large.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

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What the balance sheet cannot fix

Likely questions (2/4)

Q What do the red flags mean?

They are reasons to ask, not proof of manipulation. Four flags are raised, three tests are clean and two cannot be run on the published statements – that is why information rights come first.

Q Why is ROCE so low?

ROCE is 3.5 % before tax against a 7.70 % after-tax benchmark: the margin is only 2.8 %, and inventories of 42.9 % of revenue and a cash cycle 77 days longer than Caterpillar's tie up capital.

Q Why is a strong balance sheet not a return?

An equity ratio of 53.5 % and a net financial position of €249 m limit the risk, but do not earn the cost of capital. They would cover a shortfall like 2025 (−€232 m) for about 1.1 years.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

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Questions on method

Likely questions (3/4)

Q Why pre-tax ROE?	The effective tax rate was 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), with no tax reconciliation published – the after-tax trend says little.
Q Why Caterpillar as the peer?	The world's leading maker of construction and mining equipment – the core market of Liebherr's largest sector (63.3 % of revenue). We compare ratios only, never € and $ figures.
Q Why a US industry benchmark?	Damodaran publishes consistent data for 105 listed machinery companies. It is a reference level, not Liebherr's own cost of capital.
Q R&D expensed, yet €317 m capitalised?	Own work capitalised is self-built assets and rental machines (IAS 16), not development costs (IAS 38), which are never capitalised.
Q Why days on a revenue basis?	Liebherr reports no cost of sales, so DIO and DPO use revenue for both groups – comparable, but not the textbook formula.

Source: Report §4.2, §5.3, §5.7 and §6.3; Excel workbook, sheets C2 and D3; Caterpillar Inc. (2026)

More questions

Likely questions (4/4)

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Q What is the sustainable growth rate?	ROE × (1 − payout ratio) = 2.4 % in 2025. Faster growth needs better returns, lower capital intensity or outside financing.
Q Does the advice differ by investor?	Risk-averse: equity unsuitable, a loan may fit better. Long-term: not yet. Risk-seeking: only at a price that reflects the risk – without influence as a minority.
Q Why is the pre-tax margin above the operating margin?	3.0 % against 2.8 %: the finance result was positive in 2025 (+€24 m), partly fair value gains on securities (€55 m).
Q Which DSO do you quote?	73.8 days on average balances (ratio table) or 77 days on year-end balances (strategy chapter) – say which basis.
Q What is not on the balance sheet?	Guarantees for customer financing – the volume is not disclosed in any year, so receivables understate the credit exposure.

Source: Report §4.3, §5.3, §5.5, chapter 8 and Tables G2, H2; Excel workbook, sheets A1, D3 and D8

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